

TECH HORIZON 2.0

Integrated Hackathon Proposal

Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots

Decentralized Intelligence for Faster, Safer and More Resilient Smart Warehouses

Executive Summary

Modern warehouse automation increasingly depends on fleets of Autonomous Mobile Robots (AMRs) to move products between storage, pickup and drop-off locations. Conventional centralized fleet architecture places coordination and decision-making at a central controller. As fleet size and operational complexity increase, this can introduce decision bottlenecks, network dependency, latency, congestion at shared spaces and a single point of failure.

This proposal presents an **Edge-AI based distributed fleet coordination architecture** in which each AMR can perceive its environment, make local decisions, communicate directly with peer robots, negotiate conflicts, dynamically re-plan and continue essential coordination when central connectivity is disrupted.

The hackathon prototype will demonstrate **3–5 AMRs** in a warehouse simulation or controlled AMR setup with narrow aisles, overlapping tasks, dynamic obstacles and simulated network disruption. The validation targets zero inter-robot collisions, successful conflict resolution and dynamic rerouting, continued operation during disruption, and a target of **≥20% reduction in total task completion time** versus a basic stop-and-wait strategy, subject to experimental validation.

Project Positioning: We are not building a new AMR. We are developing a distributed intelligence and coordination layer that can run at the robot edge and integrate with existing AMR platforms.

1. Problem Statement

Centralized warehouse robotics systems can become less effective as the number of robots and the complexity of the environment increase. When multiple robots encounter the same constrained area, they may have to wait for a central controller to resolve the conflict. Network latency or Wi-Fi dead zones can further delay decisions, while loss of the central service can pause fleet operations.

- **Centralized bottleneck:** a single decision point can limit responsiveness as fleet size grows.
- **Network dependency:** communication delays and connectivity gaps can affect real-time coordination.
- **Single point of failure:** central-controller failure can interrupt fleet-level operation.
- **Choke-point congestion:** robots may stop and wait when their paths overlap in narrow aisles.
- **Limited local adaptability:** task allocation and routing may not respond quickly to local conditions.

Representative scenario: Two robots reach a narrow aisle simultaneously. Instead of independently resolving the conflict, both may wait for a central command, creating congestion and avoidable lost time.

The core challenge is therefore to move time-critical coordination closer to the robots while retaining optional centralized monitoring and high-level management.

2. Proposed Solution

The proposed system distributes intelligence across the fleet. Each AMR follows the local decision loop:

PERCEIVE → DECIDE → COMMUNICATE → COORDINATE → ACT

3. End-to-End System Workflow

Stage	Function
Perception	LiDAR, cameras and IMU observe the environment and robot state.
Edge AI processing	Lightweight detection and short-term prediction run locally.
Localization	SLAM + IMU + odometry provide a robust local pose estimate.
Local decision	The robot plans a path and performs local collision avoidance.
Peer coordination	Robots exchange position, velocity, task, path, intended direction, obstacle and priority information.
Conflict resolution	Potential path overlap is detected; priority and ETA are compared.
Negotiation	Robots adjust speed or select an alternative route.
Action / confirmation	The agreed plan is executed and the fleet continues.
Dynamic response	New obstacles trigger local broadcasts and fleet re-planning.

Example: Robot A detects a blocked aisle and broadcasts the obstacle information. Robot B receives the update, and both robots locally re-plan so the fleet can continue without waiting for a central command.

4. Distributed Coordination Mechanism

Robots periodically broadcast relevant state and intent information to nearby peers:

- Position and velocity
- Current task
- Planned path
- Intended direction
- Obstacle information
- Priority level

Local conflict-resolution sequence

- Detect potential path overlap.
- Compare priority and estimated time of arrival (ETA).
- Negotiate by adjusting speed or selecting an alternative route.
- Confirm the resulting plan and continue execution.

The central design principle is distributed decision-making with explicit conflict resolution and deadlock prevention rather than simple stop-and-wait behavior.

5. Technical Architecture

Edge Intelligence

- Perception: lightweight object detection using YOLO variants, semantic cues and LiDAR-based obstacle sensing.
- Localization: SLAM combined with IMU and odometry fusion.
- Path planning: A* / D* Lite with dynamic re-planning and velocity profiling.
- Multi-agent coordination: priority negotiation, distributed task allocation and conflict detection.
- Edge targets: NVIDIA Jetson-class hardware, industrial ARM boards or equivalent platforms, with models optimized for latency and power.

Communication and Infrastructure

Technology	Role
ROS2 / DDS	Robot-to-robot and robot-to-edge communication framework.
WebRTC DataChannel	Low-latency peer data exchange option.
gRPC	Service/API communication between software components.
Wi-Fi 6 / 5G / TSN	Candidate network infrastructure depending on deployment.

6. End-to-End System Layers

Layer	Responsibilities	Resilience / Role
Device Layer	Edge inference, local planning and peer-to-peer low-latency communication.	Continues operating if the server becomes unavailable.
Edge Server Layer	Global map, model management and monitoring; optional central task allocation and analytics.	Supports fleet-level management without being required for every local decision.
Optional Cloud	Long-term analytics, dashboards and OTA updates.	Provides extended management and data capabilities.

7. Why Distributed Edge-AI?

Dimension	Centralized Approach	Distributed Edge-AI
Decision-making	Central bottleneck; single point of failure	Local decisions; no single decision bottleneck
Latency	Higher network dependency; greater latency	Low-latency local response with peer coordination
Resilience	Operations can be affected during outages	Continues operating during network disruption
Scalability	Limited local autonomy as fleet complexity grows	Fleet can scale through peer negotiation

Priority outcomes: LOWER LATENCY • HIGHER RESILIENCE • BETTER FLEET UTILIZATION

8. Hackathon Demonstration Plan

- Deploy 3–5 AMRs in a warehouse simulation or controlled AMR environment.
- Assign overlapping pickup/drop tasks to create competing route requirements.
- Introduce a narrow aisle or choke point where robot paths can conflict.
- Allow robots to detect the conflict, exchange state and negotiate priority.
- Introduce a dynamic obstacle that blocks an aisle and observe peer-to-peer broadcast and re-routing.
- Simulate network disruption and verify that local robot coordination continues without central approval.
- Record task completion time, waiting time, collision status and continuity during disruption.

9. Success Criteria and Validation

Metric	Target / Validation
Inter-robot safety	Zero inter-robot collisions during the demonstration.
Conflict resolution	Successful local resolution of overlapping paths.
Dynamic re-routing	Successful fleet re-routing around a newly introduced obstacle.
Network resilience	Continued operation during simulated network disruption.
Efficiency	Target $\geq 20\%$ reduction in total task completion time versus a basic stop-and-wait strategy; expected and to be validated experimentally.

10. Expected Impact

- Real-time coordination between robots.
- Reduced congestion and waiting at shared spaces.
- Improved fleet utilization.
- Lower dependence on a continuously available central controller.
- Resilient operation during network disruption.

The architecture moves time-critical coordination closer to the robots, enabling local responses while retaining optional central monitoring and high-level management.

11. Future Roadmap

- Federated learning for distributed model improvement.
- Digital twin integration for warehouse simulation, testing and optimization.
- Multi-warehouse coordination.
- Human-robot collaboration.

12. Conclusion

This project proposes a shift from instruction-dependent warehouse robots toward a fleet of autonomous agents that can perceive, negotiate, adapt and act locally. By combining Edge AI, peer-to-peer communication, distributed conflict resolution and dynamic re-planning, the system aims to reduce latency, improve resilience and make more effective use of the available robot fleet.

Distributed by design. Intelligent at the edge. Autonomous as a fleet.

Core Project Positioning: We are not building a new AMR; we are building the distributed intelligence and coordination layer that can run at the robot edge and integrate with existing AMR platforms.